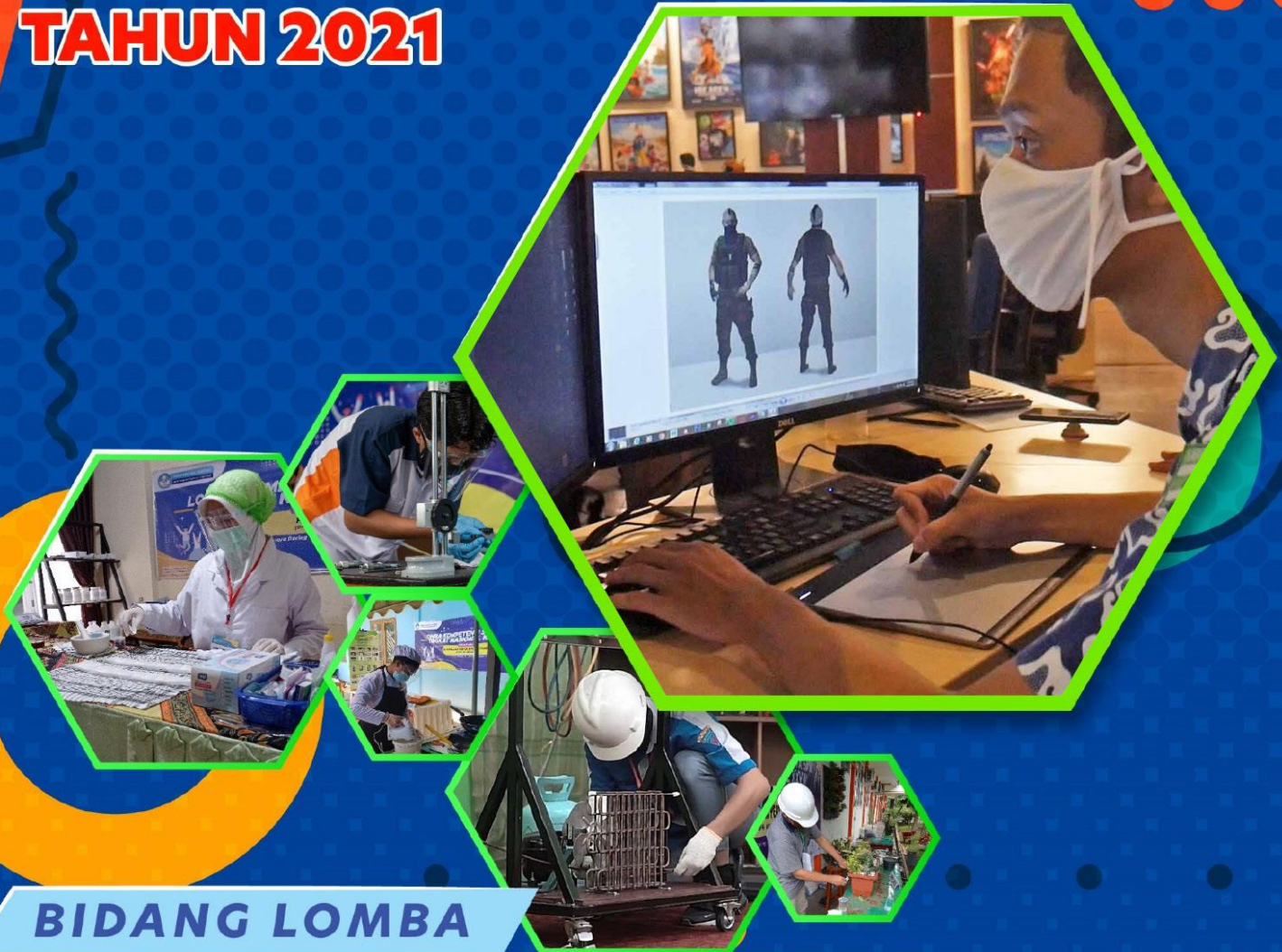




Puspresnas
Pusat Prestasi Nasional

KISI-KISI DAN SOAL-SOAL

**LOMBA KOMPETISI SISWA (LKS)
TINGKAT NASIONAL XXIX
TAHUN 2021**



BIDANG LOMBA

Teknik Perancangan Model 3 Dimensi
3D Game Art

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1. Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to LKS.

- Introduction/overview
- Short description of project and tasks
- Instruction to the Competitor

2. Introduction

The goal of this competition is to develop assets for the newest version of the game

a. Game Background

Explanation about the game background story

The alien suddenly appears on earth and attacking the human race with their robots. It has caused great suffering and pain. At this moment every nation had created superheroes to defend their country. In Indonesia, many superheroes came to save their city, one of the superheroes is "Captain Garuda". He leads a team of superheroes in Indonesia to protect people from the alien's Robot. Could Captain Garuda and his team defend Indonesia from the vicious attack?

b. The Brief

Through the competition, you will be creating Asset for the game (please choose one of these two assets):

- Major Asset: **Captain Garuda**
 - Description for major asset
 - (The armour worn is made of titanium, the result of the latest research, called palladium, combined with soft fibers that are heat-resistant and attack toxic substances and are able to withstand nuclear explosions)
 - garuda logo
 - 1st class armour
 - flying wings
 - unique saber sword
- Minor Asset: **Aliens Robots**
 - Description for minor asset
 - (aliens have the power to destroy thanks to the fusion of robotic biotechnology implanted in their bodies)
 - half-machine body
 - high crushing power
 - medium-range automatic missile weapon

c. Design project

Organic

Image of Captain Garuda



Aliens Robots

Must have complete information about the images



d. Schedule Overview

Results of this Competition will including the following:

Day	DELIVERABLE	DESCRIPTION
1	Modeling	- Sculpt the character (Captain Garuda or Aliens Robot) - o Must be high poly and view presentation must be from the front, left, right, 3/4
2	Modeling and UV unwrapping	- Finish the Character (Captain Garuda or Aliens Robot) - o The model with a maximum of 50.000 triangles - Unwrap character - Unwrap props
3	Texturing and Presentation	- Texture the Character (Captain Garuda or Aliens Robot) - o 4096 x 4096px texture map - o Ambient occlusion, normal, and opacity maps - o Diffuse/Metal/Rough PBR - Texture props - o 1024 x 1024px texture map - o Ambient occlusion, normal, and opacity maps - o Diffuse/Metal/Rough PBR - Presentation - o The character must be posed - o The character must be a presentation from the front, left, right, and 3/4 view - o The character must be a presentation from the front, left, right, and 3/4 view with Wireframe

NOTE – ALL ASSETS ARE TO BE PLACED IN A FOLDER ON THE SERVER

3. Description of project and task

a. Day 1 - Sculpting

Day 1 Brief

Example for the brief

You will be creating assets for Captain Garuda, and **you need to sculpt the character (Captain Garuda or Aliens Robot)**. Following this, you need to create a 3D sculpt and dummy model for the competition.

The committee will provide:

- The concept art for organic character and hard surface character
- The base character model (can create from zero)
- Presentation template

Day 1 Deliverables

By the end of today, you will need to deliver :

1. 3D Sculpt for the character such as the concept art
2. 3D Sculpt Presentation view from the front, Side, 3/4 for the organic character.
or
1. The base model for props
2. Base model presentation view from the front, Side, 3/4 for the hard surface character.

NOTE – ALL ASSETS ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 1 General Guidelines

Description

- 3D Sculpt for the organic character filename structure must be like **“CHR_NameOrganicCharacter_Sculpt”**
- The base model for a hard surface character structure must be like **“CHR_NameHardsurfaceCharacter_Dummy”**
- the file extension must be support for Maya, 3dmax, blender, ZBrush, or related 3D software support 3D sculpt i.e. **.fbx, .obj, .ztl, .zbr**

b. Day 2 – Modeling and unwrap UV

Day 2 Brief

Example for the brief

The 3D sculpt for character or base model weapon must be complete. It is not important in which order you complete the models as long as they are both ready for marking by the end of today. The 3D Sculpt character should be retopology and must remain within the 50.000 triangle limit.

The base model of the props must to be complete today. Both of them must be UV unwrapped.

Day 2 Deliverables

- The Character (Captain Garuda or Aliens Robot) – Maximum 50.000 triangles
- Props – Maximum 10.000 triangles
- Both of them must be UV unwrapping

NOTE – ALL ASSETS ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 2 General Guidelines

Description

1. Maya, Max, or Blender Scene Organisation – only relevant assets in the scene
2. Object pivot must be in zero grid coordinate (0,0,0)
3. The pivot must be below the object
4. Unit in Scene must be 10cm/grid
5. Topology and edge loop distribution for character and props. Consider areas that require deformation
6. Both 3D models conform to edge flow that accentuates the represented asset.
7. (major/minor) Assets can be understood in silhouette only
8. No N-gons are present on the assets.
9. No flipped normal present on the assets.
10. Chamfered edges on any 90 degrees on the assets
11. The model utilizes 90+% of the triangle budget on the assets.
12. Model is within triangle budget
13. Character filename structure **“CHAR_NameCharacter_MDL”**
14. Props filename structure **“PROP_NameOfWeapon_Dummy”**
15. the file extension must be support for Maya, 3dmax, blender i.e.**.fbx, .obj**

c. Day 3 - Texturing and Presentation

Day 3 Brief

Example for the brief

Texture the Organic Character or Hard surface, character texture sheets of 4096 x 4096px (Characters) and 1024 x 1024px (Props). Do not make the assets share 1 sheet.

Make the most out of each sheet – note that marks are awarded for the effective use of the texture sheets.

Create texture for each asset, though you may also paint detail and create maps in 3d painting software such as a substance painter, photoshop, Affinity photo, substance designer, or 3dCoat. Think about the variety of materials that may be used, for example, metal, plastic, skin, fiber, etc.

Make sure to follow the look and feel of the reference images.

Day 3 Deliverables

1. texture map for the Organic Character or Hard surface
 - Diffuse map
 - Metal map
 - Roughness map
 - Ambiance Occlusion (AO) map
 - Normal map
 - Opacity map
2. 2 texture sheets
 - Character – 4096 x 4096px
 - Props – 1024 x 1024px
3. Final 3D model for the character and props with the final texture
4. Organic or Hard surface must be posed for presentation
5. Organic or Hard surface must be a presentation from perspective view (3/4 front and 3/4 back)

NOTE – ALL ASSETS ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 3 General Guidelines

Description

1. Aspect Ratio of texture sheets are correct – **4096x4096(major) & 1024x1024(minor)**
2. PSD saved as PNG format and applied to the node
3. UV Tidiness and Efficiency - character
4. Smooth and even UV shells
5. Texel density – the size of polygons relative to the pixel resolution – Character and Props
6. Group shells with similar colors together
7. UV coordinates exported to PS or Substance
8. UV Tidiness and Efficiency – Props
9. No overlapping UVs (except for mirror/repeat)
10. UVs utilize rotation to maximize space
11. No obvious pixelation on the surface
12. No obvious stretching of textures on the surface
13. Colour and Tone represents the base color of the material
14. Surface Textures - describes materials correctly
15. The texture looks seamless on model
16. The texture is consistent with an art style
17. Specular or Metal map created
18. Normal map created
19. Roughness or smoothness map created
20. A variety of physical materials have been represented
21. Appropriate use of Opacity map created for transparency
22. Ambient Occlusion map created
23. 2 or more base colors have been used on the texture sheet
24. Height or Displacement map created
25. Smart masks used in Substance Painter
26. Evidence of variation in PBR textures (settings)
27. For character, the filename must be **“CHAR_nameCharacter_ FINAL”**

28. Texture filename for character must be :

- Diffuse = “CHAR_nameCharacter_DIFF”,
- AO = “CHAR_nameCharacter_AO”,
- Metal = “CHAR_nameCharacter_MTL”
- Rough = “CHAR_nameCharacter_RGH”
- Opacity = “CHAR_nameCharacter_OPT”
- Normal = “CHAR_nameCharacter_NRM”

4. Instructions to the Competitor

Create a folder on the server provided by the committee

The structure folder example must be like in the below:

RootFolder must be named “**your sub Provice_YourName_YourSchool**”

example

Folder name : **Surakarta_agusKuncoro_SMK7Kudus**

The folder must be “**Year + Month + Day**” the example folder name must be “**20210201**”

Year = 2021

Month = 02

Day = 01

Every Final progress must upload in that folder

Internet Access

- Need internet access

Remember THE STANDARDS SPECIFICATION...!!!



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